Topics in Error-Correcting Codes

Notes by Rohan Goyal

MIT — 6.S953

Lecture 15: Bipartite Perfect Matching in NC

In this lecture we go through the breakthrough paper *Bipartite Matching is in NC* (2026) by Abhranil Chatterjee, Sumanta Ghosh, Rohit Gurjar, Roshan Raj, and Thomas Thierauf. The paper gives a deterministic NC algorithm for bipartite matching, resolving a major open problem in algorithm design that had stood for decades. We will present the perfect-matching decision algorithm through local profiles and subspace designs.

The idea of using subspace designs to construct algorithms via threshold computation, together with the view of containment through nonzero witnesses used here, is due to Joshua Brakensiek, Yeyuan Chen, Manik Dhar, and Zihan Zhang in *From Random to Explicit via Subspace Designs With Applications to Local Properties and Matroids*.

Local Profiles: A Different Notion of Containment

Recall that an *r-local coordinate-wise linear profile* is a tuple $\mathcal{V} = (V_1, \dots, V_n)$ of linear maps

$$V_i : \mathbb{F}_q^r \longrightarrow \mathbb{F}_q^r.$$

Only the allowed spaces $\ker V_i$ matter. As in Lecture 13, our codes are $\mathbb{F}_q$-linear subspaces $\mathcal{C} \subseteq (\mathbb{F}_q^s)^n$, and each map is applied separately to the s scalar entries of a vector symbol. All dimensions are over $\mathbb{F}_q$.

In Lecture 13, containment required *linearly independent* codewords $c_1, \dots, c_r$. Today we use a weaker definition: *at least one of these codewords must be nonzero*.

Definition 1 (Containment for this lecture)**.** The code $\mathcal{C}$ *contains* $\mathcal{V}$ if there exist $c_1, \dots, c_r \in \mathcal{C}$, not all zero, such that

$$V_i \begin{pmatrix} c_{1,i} \\ \vdots \\ c_{r,i} \end{pmatrix} = 0 \qquad (i \in [n]).$$

The displayed matrix has r rows and s columns.

We use the potential $\Phi_W((V_1, \dots, V_n), \alpha)$ defined in Lecture 13, abbreviated $\Phi_W(\mathcal{V}, \alpha)$. Strict negativity conditions range over *nonzero* subspaces W, since $\Phi_{\{0\}} = 0$.

Negative potential rules out containment

Recall that $\mathcal{C}$ is a τ-subspace-design code if, for every subspace $\mathcal{A} \subseteq \mathcal{C}$ of dimension d,

$$\frac{1}{n}\sum_{i=1}^{n} \dim \mathcal{A}_i \le \tau(d)d, \qquad \mathcal{A}_i = \{c \in \mathcal{A} : c_i = 0\}.$$

Proposition 2 (Noncontainment from a subspace design)**.** *If $\mathcal{C}$ is a τ-subspace-design code and*

$$\Phi_W\big(\mathcal{V}, \tau(\dim W)\big) < 0 \qquad \textit{for every } 0 \neq W \le \mathbb{F}_q^r,$$

then $\mathcal{C}$ does not contain $\mathcal{V}$.

Proof. Suppose $c_1, \dots, c_r$ witness containment, and let $\mathcal{A} = \operatorname{span}\{c_1, \dots, c_r\}$ have dimension $d \ge 1$. Choose a basis $b_1, \dots, b_d$ of $\mathcal{A}$ and write

$$\begin{pmatrix} c_1 \\ \vdots \\ c_r \end{pmatrix} = A \begin{pmatrix} b_1 \\ \vdots \\ b_d \end{pmatrix}, \qquad A \in \mathbb{F}_q^{r\times d} \text{ injective.}$$

Set $W = \operatorname{im} A$. At coordinate i, every row of $V_i A$ gives a linear combination of the b_j whose ith symbol is zero. Since A identifies $\mathbb{F}_q^d$ with W, these coefficient vectors span a space of dimension $\dim W - \dim(W \cap \ker V_i)$. They lie in a space isomorphic to $\mathcal{A}_i$, so

$$\dim W - \dim(W \cap \ker V_i) \le \dim \mathcal{A}_i.$$

Averaging and applying the design bound gives $\Phi_W(\mathcal{V}, \tau(d)) \ge 0$, a contradiction. □

Unlike the independent-witness definition, a witness may now span any dimension from 1 to r. This is why the condition must hold on every nonzero subspace W.

Positive potential forces containment

Proposition 3 (Containment by counting dimensions)**.** *Let $\mathcal{C} \subseteq (\mathbb{F}_q^s)^n$ be any linear code of dimension k and rate $R = k/(sn)$. If some $0 \neq W \le \mathbb{F}_q^r$ satisfies*

$$\Phi_W(\mathcal{V}, R) > 0,$$

then $\mathcal{C}$ contains $\mathcal{V}$.

Proof. Put $d = \dim W$ and choose an injective matrix $A : \mathbb{F}_q^d \to \mathbb{F}_q^r$ with image W. Take d unknown codewords $b_1, \dots, b_d \in \mathcal{C}$, with kd message coefficients in total, and set $(c_1, \dots, c_r)^{\mathsf{T}} = A(b_1, \dots, b_d)^{\mathsf{T}}$.

At coordinate i, the condition $V_i A(b_{1,i}, \dots, b_{d,i})^{\mathsf{T}} = 0$ imposes at most $s\big(\dim W - \dim(W \cap \ker V_i)\big)$ scalar linear equations. Thus the space of solutions has dimension at least

$$kd - s\sum_{i=1}^{n}\big(\dim W - \dim(W \cap \ker V_i)\big) = sn\,\Phi_W(\mathcal{V}, R) > 0.$$

A nonzero solution exists. Since A is injective, its image is a tuple of codewords that is not all zero. □

This implication is deterministic and holds for every code of rate R; no randomness or subspace-design assumption is needed.

Evaluating a Profile's Threshold with FRS Codes

The two propositions suggest the following threshold for our new notion of containment:

$$\boxed{R_{\mathcal{V}} = \min_{0 \neq W \leq \mathbb{F}_q^r} \frac{\sum_{i=1}^{n} \big(\dim W - \dim(W \cap \ker V_i)\big)}{n \dim W}.} \tag{1}$$

In particular,

$$R > R_{\mathcal{V}} \quad \Longleftrightarrow \quad \Phi_W(\mathcal{V}, R) > 0 \text{ for some } W \neq 0.$$

This is the threshold associated with *nonzero* witnesses; the threshold in Lecture 13 used independent witnesses.

For an s-folded Reed–Solomon code of rate R, Lecture 13 gives, for $1 \leq d \leq r \leq s$,

$$\tau(d) \leq \frac{sR}{s-d+1} \leq R + \frac{R(r-1)}{s-r+1}.$$

Given $\eta > 0$, taking s sufficiently large makes $\tau(d) \leq R + \eta$ for every $d \leq r$. For example, it is enough that $s \geq r$ and $s - r + 1 \geq (r-1)/\eta$. The two propositions now give

$$R > R_{\mathcal{V}} \quad \Longrightarrow \quad \mathcal{C}_{\mathrm{FRS}} \text{ contains } \mathcal{V}, \tag{2}$$

$$R + \eta < R_{\mathcal{V}} \quad \Longrightarrow \quad \mathcal{C}_{\mathrm{FRS}} \text{ does not contain } \mathcal{V}. \tag{3}$$

Thus containment in an explicit FRS code tests the threshold, with an arbitrarily small slack controlled by the folding parameter.

Why the test is in NC

Write each candidate codeword as the FRS encoding of

$$f_j(X) = \sum_{h=0}^{k-1} u_{j,h} X^h \qquad (j \in [r]).$$

The containment equations are

$$V_i \begin{pmatrix} f_1(\alpha_i \gamma^a) \\ \vdots \\ f_r(\alpha_i \gamma^a) \end{pmatrix} = 0 \qquad (i \in [n],\ 0 \leq a < s).$$

These form one homogeneous linear system $M_{\mathcal{V}} u = 0$ in the rk coefficients $u_{j,h}$. FRS encoding is injective, so

$$\mathcal{C}_{\mathrm{FRS}} \text{ contains } \mathcal{V} \quad \Longleftrightarrow \quad \ker M_{\mathcal{V}} \neq \{0\} \quad \Longleftrightarrow \quad \operatorname{rank} M_{\mathcal{V}} < rk.$$

The test is for a *nonzero* solution: the zero solution is always present. Computing matrix rank over a finite field is known to be in NC, meaning polynomially many processors and polylogarithmic parallel time. The entries of $M_{\mathcal{V}}$ are also computable in parallel. For inverse-polynomial η, the folding parameter and the linear system have polynomial size.

Running the tests for all message dimensions k in parallel locates $R_{\mathcal{V}}$ to inverse-polynomial accuracy. More precisely, let k_0 be the first dimension for which containment holds. Then

$$\frac{k_0 - 1}{sn} \le R_{\mathcal{V}} \le \frac{k_0}{sn} + \eta.$$

If containment never holds, the full-rate code gives $R_{\mathcal{V}} = 1$. The threshold is a rational number $a/(nd)$ with $1 \le d \le r$ and $0 \le a \le nd$. Distinct such values are separated by at least $1/(n^2r^2)$. Choosing s and η so that $1/(sn) + \eta < 1/(n^2r^2)$ therefore allows exact recovery by testing the polynomially many candidate values in parallel, over a field large enough for these FRS parameters.

Our goal. We will capture bipartite perfect matching as one of these threshold evaluations. The reduction will have an inverse-polynomial gap, so even the approximate threshold test suffices.

Hall's Condition as a Subspace Optimization

Let $G = ([m]_L, [m]_R, E)$ be a bipartite graph. For a set $I \subseteq [m]_L$, let $N(I) \subseteq [m]_R$ be its neighbors; for a right vertex j, let $N(j) \subseteq [m]_L$ be its neighbors. Hall's theorem says

$$G \text{ has a perfect matching} \quad \Longleftrightarrow \quad |N(I)| \ge |I| \quad \text{for every } I \subseteq [m]_L.$$

Equivalently, defining

$$h(G) = \min_{\varnothing \ne I \subseteq [m]} \frac{|N(I)|}{|I|},$$

there is no perfect matching exactly when $h(G) < 1$. Taking $I = [m]$ shows $h(G) \le 1$, so a perfect matching is equivalent to $h(G) = 1$.

Replacing subsets by coordinate subspaces

For each nonempty $I \subseteq [m]$, let

$$U_I = \operatorname{span}\{e_i : i \in I\} \le \mathbb{F}_q^m.$$

This is the space of vectors supported on I, and $\dim U_I = |I|$. For each right vertex j, define

$$\ell_j(x) = \sum_{i \in N(j)} x_i, \qquad H_j = \ker \ell_j.$$

If j is isolated, $H_j = \mathbb{F}_q^m$. Otherwise H_j is a hyperplane. Use a local map $V_j(x) = \ell_j(x)e_1$, so that $\ker V_j = H_j$. The equation $\ell_j(x) = 0$ cuts out a codimension-one

subspace of U_I precisely when $I \cap N(j) \neq \varnothing$. Hence

$$\dim U_I - \dim(U_I \cap H_j) = \begin{cases} 1, & I \cap N(j) \neq \varnothing, \\ 0, & I \cap N(j) = \varnothing. \end{cases}$$

Summing over right vertices counts each neighbor of I once:

$$h(G) = \min_{\varnothing \neq I \subseteq [m]} \frac{\sum_{j=1}^{m} \big(\dim U_I - \dim(U_I \cap H_j)\big)}{\dim U_I}. \tag{4}$$

This has the form of a profile threshold, apart from normalization by the number of local constraints.

The gap in the reduction. Hall's condition minimizes only over the coordinate subspaces U_I, whereas a profile threshold minimizes over *all* nonzero subspaces. On a general subspace, the sums ℓ_j may vanish through cancellation. For example, in the complete bipartite graph on two vertices per side, every $\ell_j = x_1 + x_2$ vanishes on $\operatorname{span}\{(1,-1)\}$, although the graph has a perfect matching. We must force the minimum to occur at a coordinate subspace.

Padding Forces a Coordinate Subspace

For each left coordinate $i \in [m]$, introduce the coordinate hyperplane

$$K_i = \{x \in \mathbb{F}_q^m : x_i = 0\}.$$

Pad the profile with *m copies of each* K_i, in addition to the m original allowed spaces H_j. Each copy is represented by the map $x \mapsto x_i e_1$. The padded profile $\widetilde{\mathcal{V}}$ has locality m and length

$$N = m + m^2.$$

For a nonzero subspace $W \leq \mathbb{F}_q^m$, put $d = \dim W$ and define its support by

$$T(W) = \{i : \text{some } w \in W \text{ has } w_i \neq 0\}, \qquad t = |T(W)|.$$

The difference $\dim W - \dim(W \cap K_i)$ is one exactly when $i \in T(W)$, and zero otherwise. The padding therefore contributes mt to the total constraint count, giving

$$NR_{\widetilde{\mathcal{V}}} = \min_{0 \neq W \leq \mathbb{F}_q^m} \frac{mt + \sum_{j=1}^{m} \big(\dim W - \dim(W \cap H_j)\big)}{\dim W}. \tag{5}$$

Lemma 4 (The minimum is attained at a coordinate subspace)**.** *The padded profile satisfies*

$$\boxed{NR_{\widetilde{\mathcal{V}}} = m + h(G).}$$

Proof. Since $W \subseteq U_{T(W)}$, we have $t \geq d$, with equality exactly when $W = U_{T(W)}$ is a coordinate subspace.

If $W = U_I$, then $t = d = |I|$. By (4), its value in (5) is

$$m + \frac{|N(I)|}{|I|}.$$

The minimum over coordinate subspaces is $m + h(G) \leq m + 1$.

If W is not a coordinate subspace, then $t \geq d + 1$ and $d \leq m - 1$. Since each $\dim W - \dim(W \cap H_j) \geq 0$, its value is at least

$$\frac{mt}{d} \geq m + \frac{m}{d} > m + 1.$$

Such a subspace cannot minimize the objective. □

The padding charges for the number of coordinates used by W, while the denominator counts its dimension. Coordinate subspaces use exactly one coordinate per dimension; every other subspace pays an extra cost.

The threshold gap and the parallel test

Since $(m + 1)/N = 1/m$, the lemma gives

$$G \text{ has a perfect matching} \iff R_{\widetilde{\mathcal{V}}} = \frac{1}{m}.$$

If there is no perfect matching, Hall gives a nonempty I with $|N(I)| \leq |I| - 1$, and therefore

$$h(G) \leq 1 - \frac{1}{|I|} \leq 1 - \frac{1}{m}, \qquad R_{\widetilde{\mathcal{V}}} \leq \frac{1}{m} - \frac{1}{mN}.$$

Thus the two cases are separated by the inverse-polynomial gap $\delta = 1/(mN)$. Choose an FRS code of length N with

$$R = \frac{1}{m} - \frac{\delta}{2}, \qquad \tau(d) \leq R + \frac{\delta}{4} \quad (1 \leq d \leq m).$$

The folding parameter can be chosen polynomially large and with the divisibility needed to make $k = RsN$ an integer. Then

$$G \text{ has no perfect matching} \implies R > R_{\widetilde{\mathcal{V}}} \implies \text{containment},$$
$$G \text{ has a perfect matching} \implies R + \delta/4 < R_{\widetilde{\mathcal{V}}} \implies \text{noncontainment}.$$